

# OPSsig Self-Evaluation Report (OPSsig-512, Performance-Optimized Implementation)

## 1 Basic Algorithm Information

| Item                           | Value                                                            |
|--------------------------------|------------------------------------------------------------------|
| Algorithm category             | Digital Signature                                                |
| Algorithm name                 | OPSsig                                                           |
| Sub-function                   | Key generation, signature generation, and signature verification |
| Implementation version         | Performance-Optimized Implementation                             |
| Submission version label       | x86 Performance-Optimized Implementation                         |
| Parameter set / security level | OPSsig-512                                                       |

## 2 Evaluation Environment

| Item                           | Value                                       |
|--------------------------------|---------------------------------------------|
| Processor model                | Intel(R) Xeon(R) Platinum 8575C             |
| Processor frequency (reported) | 3200.052 MHz                                |
| Logical cores                  | 4                                           |
| Installed memory               | 15698744 KiB (14.97 GiB)                    |
| Architecture                   | x86_64                                      |
| Operating system               | Ubuntu 22.04.5 LTS                          |
| Kernel version                 | 5.15.0-71-generic                           |
| Compiler version               | gcc (Ubuntu 12.3.0-1ubuntu1~22.04.3) 12.3.0 |
| Build tool version             | GNU Make 4.3; cmake version 3.22.1          |
| GNU time version               | time (GNU Time) UNKNOWN                     |
| AVX2 support reported by host  | yes                                         |
| Evaluation start time (UTC)    | 2026-06-29T13:56:08Z                        |
| Evaluation end time (UTC)      | 2026-06-29T13:57:14Z                        |

## 3 Functional Tests

The functional evaluation was executed through the current self-evaluation entry point `make selfeval`. The following checks passed:

- Known-answer test (KAT) generation completed successfully and wrote vectors to:  
`Test_Vectors/avx2/OPSsig-512/KAT_SIG OPSsig-512-avx2.txt`
- Signature correctness regression test completed successfully according to the captured log.
- The correctness log reports the exported API sizes shown below and ends in a pass status.

### Correctness log excerpt

```
INFO: CRYPTO_PUBLICKEYBYTES = 6720
INFO: CRYPTO_SECRETKEYBYTES = 9920
INFO: CRYPTO_BYTES = 12021
OK: sign/verify test passed (10000 iterations)
```

### KAT generation log excerpt

```
Files have been saved in ../../Test_Vectors/avx2/OPSsig-512.
```

## 4 Performance Tests

Performance measurements were collected for a fixed 64-byte message, in line with the current x86 self-evaluation guideline for digital signatures. Each core operation was sampled 1000 times. The table below reports average latency, average cycle count, and throughput.

| Operation      | Samples | Avg. ns    | Avg. cycles | Ops/s   |
|----------------|---------|------------|-------------|---------|
| Key generation | 1000    | 840180.43  | 2352104.60  | 1190.22 |
| Sign (64 B)    | 1000    | 1330687.34 | 3725513.79  | 751.49  |
| Verify (64 B)  | 1000    | 815539.51  | 2283144.51  | 1226.18 |

## Supplemental Micro-Benchmarks

| Operation                 | Avg. ns   | Avg. cycles | Ops/s      |
|---------------------------|-----------|-------------|------------|
| polyvec_matrix_expand     | 526591.75 | 1474134.05  | 1899.00    |
| poly_uniform_eta          | 47789.55  | 133645.10   | 20925.08   |
| poly_uniform_gamma1       | 122288.66 | 342215.02   | 8177.37    |
| poly_ntt                  | 8104.43   | 22500.61    | 123389.28  |
| poly_invntt_tomont        | 7094.08   | 19673.28    | 140962.55  |
| poly_pointwise_montgomery | 272.52    | 618.65      | 3669468.92 |
| poly_challenge            | 10807.62  | 30111.24    | 92527.27   |

## 5 Resource Consumption Tests

### Static Memory Footprint

Static memory usage was estimated from the loaded binary sections of `test_static_mem_OPSSig` using the `size` tool. This load-only binary links the full implementation but intentionally executes no key generation, signing, or verification logic. The section totals capture code, read-only constants, initialized global data, and uninitialized global data, which corresponds to the guideline item “basic memory usage after loading but before executing algorithm operations”.

| Metric                    | Value (Bytes) | Value (MiB) |
|---------------------------|---------------|-------------|
| Text segment (.text)      | 56860         | 0.05        |
| Read-only data (.rodata)  | 8000          | 0.01        |
| Initialized data (.data)  | 16            | 0.00        |
| Uninitialized data (.bss) | 232           | 0.00        |
| Total static footprint    | 65108         | 0.06        |

### Peak Runtime Memory

Runtime memory was measured over 100 iterations of key generation, signature generation, and verification using a fixed 64-byte message.

| Metric                                     | Value (Bytes) | Value (MiB) |
|--------------------------------------------|---------------|-------------|
| Peak RSS from <code>getrusage</code>       | 2146304       | 2.05        |
| VmPeak from <code>/proc/self/status</code> | 3051520       | 2.91        |
| VmHWM from <code>/proc/self/status</code>  | 1593344       | 1.52        |

## 6 Transmission and Storage Overheads

| Artifact               | Size (Bytes) | Source                         |
|------------------------|--------------|--------------------------------|
| Public key             | 6720         | Exported API size              |
| Secret key             | 9920         | Exported API size              |
| Signature buffer bound | 12021        | CRYPTO_BYTES                   |
| Signature test message | 64           | Fixed performance/memory input |

## 7 Implementation Notes

| Item                          | Value                                                                                                                                                                        |
|-------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Third-party dependencies      | No external third-party cryptographic library is linked by the evaluated build. The evaluation uses the system compiler and standard build utilities already recorded above. |
| Randomness generation         | Deterministic DRNG support from <code>drng.c</code> is used by the self-evaluation drivers; fixed seeds are used where reproducibility is required.                          |
| Compilation flags             | <code>-O3 -march=x86-64 -mavx2 -mtune=native -flto -fomit-frame-pointer -std=c99 -Wpedantic -Wall -Wextra</code>                                                             |
| Target ISA / portability note | x86-64 with AVX2 enabled; this version uses AVX2 intrinsics and the dedicated assembly source file <code>pointwise.S</code> .                                                |
| Optimization approach         | Aggressive optimization using <code>-O3</code> , AVX2 vectorization, x86-64 tuning, LTO, and frame-pointer omission. POPCNT is not enabled in the default evaluated build.   |

## 8 Raw Evidence Index

| Evidence item                    | Location / command                                                                                                      |
|----------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| Implementation directory         | Implementations/Optimized_Implementation/<br>OPSsig-512-avx2                                                            |
| Primary evaluation<br>command    | make -C Implementations/Optimized_<br>Implementation/OPSsig-512-avx2 selfeval                                           |
| Self-evaluation orchestrator     | Self_Evaluation/scripts/run_self_eval.sh                                                                                |
| Environment collection<br>script | Self_Evaluation/scripts/collect_env.sh                                                                                  |
| Static memory script             | Self_Evaluation/scripts/measure_mem.sh                                                                                  |
| Metadata CSV                     | Self_Evaluation/results/avx2/OPSsig-512/<br>latest/metadata.csv                                                         |
| Environment CSV                  | Self_Evaluation/results/avx2/OPSsig-512/<br>latest/environment.csv                                                      |
| Correctness log                  | Self_Evaluation/results/avx2/OPSsig-512/<br>latest/correctness.log                                                      |
| Performance CSV                  | Self_Evaluation/results/avx2/OPSsig-512/<br>latest/speed.csv                                                            |
| Static memory CSV                | Self_Evaluation/results/avx2/OPSsig-512/<br>latest/memory_static.csv                                                    |
| Runtime memory CSV               | Self_Evaluation/results/avx2/OPSsig-512/<br>latest/memory_runtime.csv                                                   |
| KAT log                          | Self_Evaluation/results/avx2/OPSsig-512/<br>latest/kat.log                                                              |
| Generated KAT vectors            | Test_Vectors/avx2/OPSsig-512/KAT_SIG OPSsig-<br>512-avx2.txt                                                            |
| Static-memory binary             | Implementations/Optimized_Implementation/<br>OPSsig-512-avx2/test_static_mem OPSsig                                     |
| Input data note                  | Fixed 64-byte message for performance and memory tests;<br>reproducible seeded DRNG for correctness and KAT generation. |